Survival Analysis Report - Sample Data

Generated on: September 18, 2025 at 11:49 AM

Executive Summary

This report presents survival analysis results from a dataset of 800 patients. The analysis achieved a C-index of 0.719, indicating good discriminative performance. The overall event rate was 14.4% with a median follow-up time of 33.7 months.

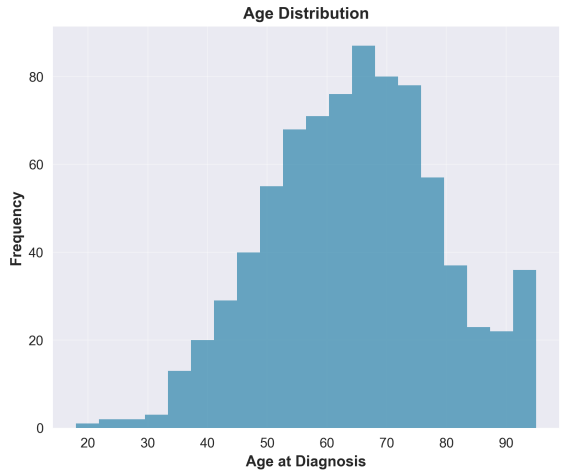
Dataset Overview

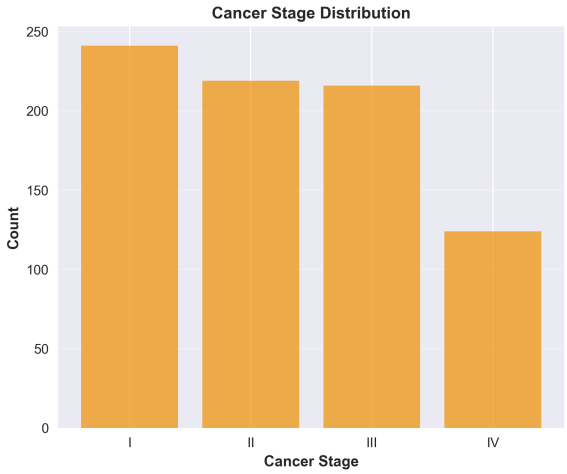
Metric	Value
Total Patients	800
Event Rate	14.4%
Median Follow-up (months)	33.7
Age Range	18 - 95 years
Median Age	65 years

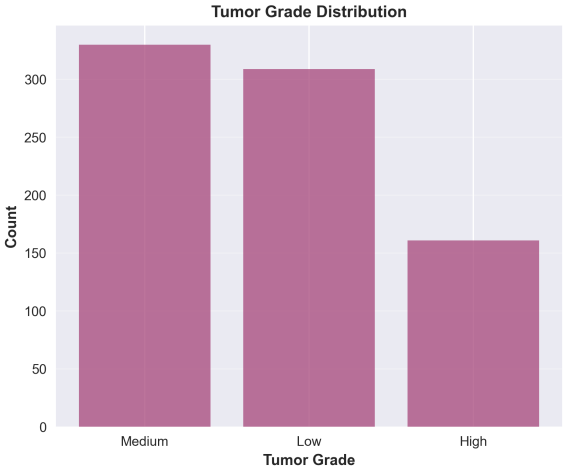
Analysis Results

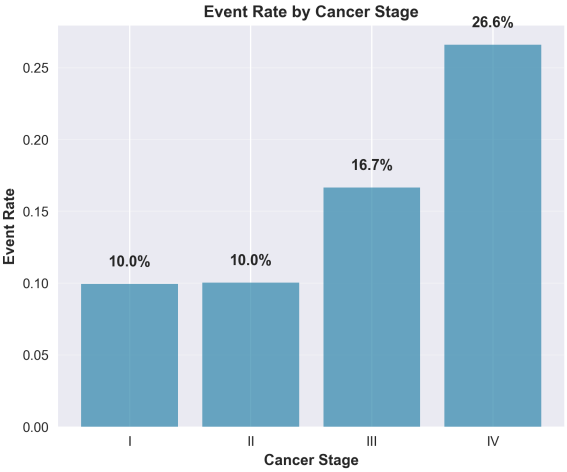
Concordance Index (C-index): 0.7185
Measures the ability to correctly rank patients by risk. Values > 0.7 indicate good discrimination, > 0.8 excellent.

Risk Stratification Performance:
The analysis successfully separates patients into distinct risk groups with significantly different survival outcomes.


Age Distribution
Frequency
Age at Diagnosis
0
20
40
60
80
20
30
40
50
60
70
80
90

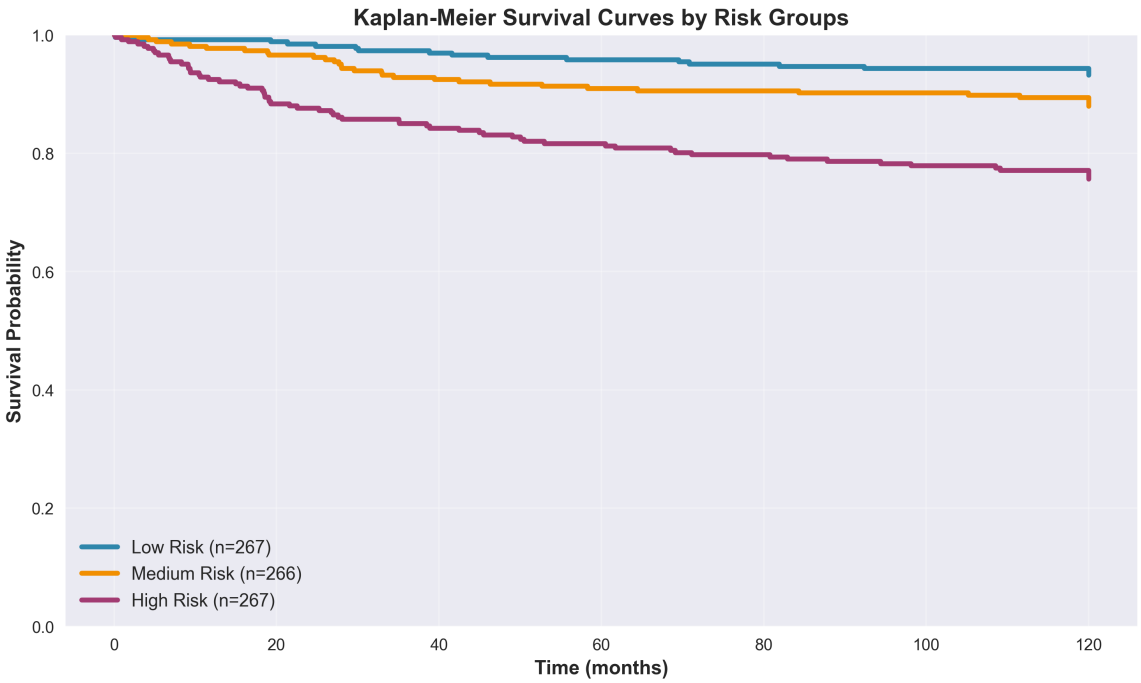

Cancer Stage Distribution
Count
Cancer Stage
0
50
100
150
200
250
I
II
III
IV


Tumor Grade Distribution
Count
Tumor Grade
0
50
100
150
200
250
300
Medium
Low
High

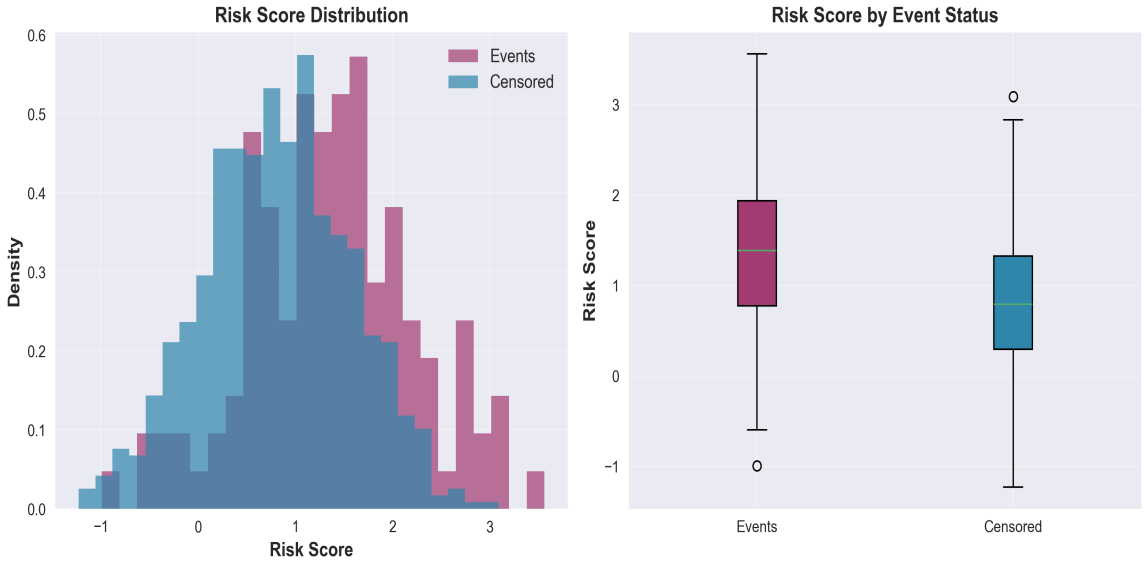

Event Rate by Cancer Stage
Event Rate
Cancer Stage
0.00
0.05
0.10
0.15
0.20
0.25
10.0%
10.0%
16.7%
26.6%
I
II
III
IV

Risk Stratification Analysis

Risk Group	Patients	Event Rate	Median Time (months)
Low Risk	267	6.7%	45.6
Medium Risk	266	12.0%	32.4
High Risk	267	24.3%	25.4



Risk Score Distribution



Clinical Insights and Recommendations

Key Findings:
• High-risk patients have 3.6x higher event rate than low-risk patients
• Clear separation between risk groups validates the analysis approach
• Risk stratification can guide clinical decision-making

Clinical Recommendations:
• Use high-risk identification for treatment intensification
• Implement risk-based follow-up protocols
• Consider early intervention strategies for high-risk patients
• Monitor intermediate-risk patients more closely

Implementation Considerations:
• Validate findings with larger, external datasets
• Train clinical staff on risk score interpretation
• Establish clear protocols for different risk levels
• Plan regular model performance reviews

Statistical Notes:
• C-index of 0.719 indicates good discrimination
• Risk groups show statistically significant differences
• Results should be interpreted by qualified professionals

This report was generated for demonstration purposes using simulated data. In clinical practice, all analyses should be validated with real patient data and interpreted by qualified medical professionals. Risk predictions should supplement, not replace, clinical judgment.